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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	Ⅰ-Ⅲ Semester(2025-26)
Faculty Name	Dr. Jayaram Boga
Student Name	Ramannagari Bhanu Priya
Roll Number	2520090171
Branch	CSIT
Course Outcome	4

Case Study Topic: CampusFlow – University Campus Resource Management System using Dijkstra's Algorithm

Work Done Summary:

Implemented Dijkstra's Algorithm in Java for campus navigation and resource management. Modeled campus facilities such as classrooms, laboratories, libraries, and administrative blocks as graph nodes connected through weighted edges representing distances between locations. Dijkstra's algorithm was used to calculate the shortest path from a source location to all other campus facilities. Implemented campus routing, facility access planning, maintenance scheduling, and resource allocation using graph-based shortest path techniques. Demonstrated shortest path computation for efficient movement of students, faculty, and maintenance staff across the campus. Compared Dijkstra's Algorithm with other shortest-path algorithms and found it suitable for campus networks with non-negative edge weights and real-time route optimization.

Implementation Details :

1. Created a Graph class using an adjacency list representation.
2. Defined an Edge class with destination node and distance (weight) fields.
3. Implemented Dijkstra's Algorithm using a Priority Queue (Min Heap).
4. Calculated the shortest distance from a source campus facility to all other facilities.
5. Modeled classrooms, laboratories, libraries, and administrative blocks as graph nodes.

6. Used weighted edges to represent distances between campus locations.
7. Displayed the shortest routes and minimum travel distances between facilities.
8. Main method created the campus graph and demonstrated shortest path analysis using Dijkstra's Algorithm.

Result : Screenshots/Output:

```
CampusFlow - University Campus Resource Management System
Using Dijkstra's Algorithm

Campus Connections:
Admin Block <-> Computer Lab = 4
Admin Block <-> Library = 2
Computer Lab <-> Classroom A = 5
Library <-> Classroom A = 3
Library <-> Seminar Hall = 6
Classroom A <-> Physics Lab = 4
Seminar Hall <-> Physics Lab = 2

Shortest Distance from Admin Block:

Admin Block -> Admin Block = 0 units
Admin Block -> Computer Lab = 4 units
Admin Block -> Library = 2 units
Admin Block -> Classroom A = 5 units
Admin Block -> Seminar Hall = 8 units
Admin Block -> Physics Lab = 9 units

Campus Route Analysis Completed Successfully.
```

Time Complexity:

- Graph Construction: **O(E)**
- Dijkstra's Traversal: **O((V + E) log V)**
- Priority Queue Operations: **O(log V)**
- Dijkstra's algorithm efficiently computes shortest paths for campus navigation and resource management systems.

GitHub Link: https://github.com/ramannagaribhanupriya/DSA-2-CaseStudies

Student Signature	**Faculty Signature**